

Information Security Network Intrusion Detection System Based on Machine Learning

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Abstract—Nowadays, more and more people are becoming aware of the security issues of computers. Traditional IDS (Intrusion Detection System) has the drawbacks of poor real-time performance and low accuracy. In response to this issue, this article adopts a machine learning(ML) based network intrusion detection algorithm. The intrusion detection module can use SVM (Support Vector Machine), deep learning, etc., to construct models. The real-time detection and response module can use the established model to monitor and analyze network traffic in real-time, in order to achieve rapid detection and response to network attacks. In the real-time performance test results, the accuracy from 08:00 to 09:00 is 0.92, the false alarm rate is 0.05, and the average detection time is 50 milliseconds; The accuracy from 09:00 to 10:00 is 0.90, the false alarm rate is 0.06, and the average detection time is 55 milliseconds. This article studies a machine learning based method for detecting and identifying network attack behaviors, which is beneficial for improving the level of network security defense.

Keywords—information security network intrusion detection system, machine learning, average detection time, real-time monitoring

I. INTRODUCTION

Nowadays, many people are becoming aware of the security issues of computers. Traditional IDS has shortcomings in both real-time performance and accuracy, so researching effective IDS is particularly important. This article mainly studies network intrusion detection technology based on ML technology. ML algorithms can extract rules and patterns from a large amount of data, thereby more accurately detecting possible attacks and responding quickly to them.

This article first provides the background of system design, points out the urgency of information security network intrusion detection today, and then conducts network construction. Finally, in the feature extraction module, the features can include information such as source IP (Internet Protocol) address, destination IP address, port number, protocol type, etc.

II. RELATED WORK

With the increasing complexity and concealment of network attack methods, traditional security protection methods are outdated. This article effectively improves the security of networks by automatically learning and identifying

different types of attacks. Zong Xuejun explored IDS for industrial control networks [1]. Hu Xiangdong explored the industrial Internet intrusion detection method based on capsule network [2]. Zhang Pengran studied network information security [3]. Li Lusheng studied campus network information security technology [4]. Mohammadpour L explored the average convolutional layer of intrusion detection systems [5]. Dong R H explored a network intrusion detection model based on multivariate correlation analysis [6]. Liu Z studied the application of deep random forest models in network intrusion detection [7]. Aryeh F L studied the application of on-demand network dataset creation for intrusion detection systems [8]. Olasehinde O O studied a stacked integrated intrusion detection method for information system protection [9]. Lakhno V studied the characteristics of network attacks in information security analysis tasks [10]. During the research process, some scholars still face the problems of poor model interpretability and large computational complexity.

In the current internet environment, intrusion detection for information security is a very important topic. With the rapid development and popularization of the Internet, the amount of data on the network has increased dramatically, bringing a variety of security risks to the network. Network intrusion refers to unauthorized users or programs invading computer systems or networks through various methods, stealing, modifying, or destroying data and resources within the computer. For enterprises, organizations, and individual users, cyber-attacks can cause property damage, information leakage, and even damage to personal privacy, posing a great threat to network security and data integrity. Machine learning models have the ability to adapt and learn, and can continuously optimize and adjust models based on changes and evolution in network attack behavior. By rapidly processing and analyzing massive amounts of data, real-time network traffic can be monitored and detected, and attacks can be detected and responded to in a timely manner. By integrating deep learning, reinforcement learning, and other methods, complex models can be established for various intrusion behaviors, achieving multi-level and multi angle intrusion detection [11-12].

III. METHOD

A. Construction of Deep Reinforcement Learning Network

1) *Step 1. Network traffic data preprocessing:* Clean and preprocess the original information security network traffic data.

2) *Step 2. Deep neural network architecture design:* Construct a deep neural network with multiple hidden layers, design appropriate excitation functions and optimization algorithms, and achieve efficient processing of high-dimensional network services.

3) *Step 3. Reinforcement learning algorithm integration:* On this basis, combined with reinforcement learning algorithms such as deep decision strategy gradients, automated identification and real-time response to network attacks are achieved, enhancing the intelligence and adaptability of the system.

4) *Step 4. Model training and optimization:* On this basis, the established deep reinforcement learning network is trained using business data with existing labels, and the optimal algorithm is used to modify the model parameters, thereby improving the performance and generalization ability of the model.

5) *Step 5. Real-time monitoring and feedback:* On this basis, the constructed deep reinforcement learning network is utilized.

B. Module Design of Information Security Network Intrusion Detection System

1) *Data collection module:* This module is responsible for collecting raw data on the network, such as network communication, logs, etc. These data may be from network devices such as firewalls, routers, or host records, etc. Data collection requires ensuring effective and timely access to network activities.

2) *Data preprocessing module:* Cleaning, denoising, and formatting the collected data provide a basis for the next step of feature extraction and model training. During the preprocessing process.

3) *Feature extraction module:* The features can be extracted from the preprocessed data for model training.

$$F1 = 2 (Precision \times Recall) / (Precision + Recall) \quad (3)$$

Among them, Precision is the accuracy rate and Recall is the recall rate.

The threshold formula for traffic anomaly detection:

$$Threshold = \mu + k \sigma \quad (4)$$

Among them, Threshold is the anomaly detection threshold, μ is the mean of traffic statistics, σ is the standard deviation of traffic statistics, and k is a constant used to adjust the threshold.

IV. RESULTS AND DISCUSSION

A. Characteristic Engineering Experiment

1) *Experimental objective:* To compare the effectiveness of various feature design schemes in IDS.

These features can include information such as the number of registrations and file access frequency recorded by the host. When extracting features, it is necessary to take into account characteristics such as high dimensionality and diversity.

4) *Intrusion detection module:* A feature extraction based intrusion detection model can be established by such as decision trees, SVM, and deep learning.

5) *Model training and optimization module:* On this basis, this article adopts a multi-objective optimization algorithm based on fuzzy clustering.

6) *Model evaluation and performance monitoring module:* The established model can be evaluated, including in the training and testing sets, and applied to real data. At the same time, performance degradation and abnormal conditions of the system determined by monitoring its operational status.

7) *Real-time detection and response module:* The system design requires real-time and high efficiency to achieve rapid identification and response to various attacks in the network environment.

8) *Alarm and logging module:* When intrusion behavior is detected, it can automatically alarm and record the corresponding log information for further analysis and processing. Alarm methods include email, SMS, instant messaging, etc.

Detection rate DR:

News:

$$DR = TP / (TP + FN) \quad (1)$$

Among them, TP is the true example and FN is the false counterexample.

The FPR formula for false alarm rate:

$$FPR = FP / (FP + TN) \quad (2)$$

Among them, FP is a false positive example and TN is a true negative example.

F1 score formula:

$$F1 = 2 (Precision \times Recall) / (Precision + Recall) \quad (3)$$

2) *Experimental process:* Multiple methods such as feature selection and dimensionality reduction were used to model it and evaluate its performance.

3) *Feature free engineering:* Based on this, a machine learning model was established using existing basic features.

4) *Feature selection:* This article intends to use methods such as chi square test to select the most relevant features and establish a model.

5) *Feature transformation:* Using polynomial features and other feature transformation techniques to transform the original features, and constructing a new model using the transformed features.

The experimental results of feature engineering are shown in Table 1. The performance of feature selection (chi square test) is excellent in all aspects.

TABLE I. RESULTS OF CHARACTERISTIC ENGINEERING EXPERIMENTS

Experimental number	Characteristic engineering method	Accuracy rate	Recall rate	F1 score
1	Feature-free engineering	0.85	0.81	0.81
2	Feature selection (chi-square test)	0.98	0.86	0.86
3	Dimension reduction (principal component analysis)	0.89	0.84	0.84
4	Feature transformation (standardization)	0.86	0.82	0.83

B. Algorithm Comparison Experiment

1) *Experimental objective*: To compare various machine learning methods.

2) *Experimental stage*: Establish models using, and compare their performance.

The experimental results of algorithm comparison are shown in Table 2. The decision tree has an accuracy rate of 0.85, a recall rate of 0.80. Random Forest has an accuracy rate of 0.90.

TABLE II. RESULTS OF ALGORITHM COMPARISON EXPERIMENT

Algorithm	Accuracy rate	Recall rate	F1 score	Precision rate
Decision tree	0.85	0.80	0.82	0.83
Random forest	0.90	0.85	0.87	0.88
Support vector machine	0.88	0.83	0.85	0.86
Neural network	0.92	0.88	0.90	0.91

C. Model Fusion Experiment

1) *Experimental objective*: To explore the role of model fusion technology in IDS.

2) *Experimental stage*: Using ensemble learning methods such as Bagging (Bootstrap Aggregating) and Boosting, multiple models are fused and evaluated.

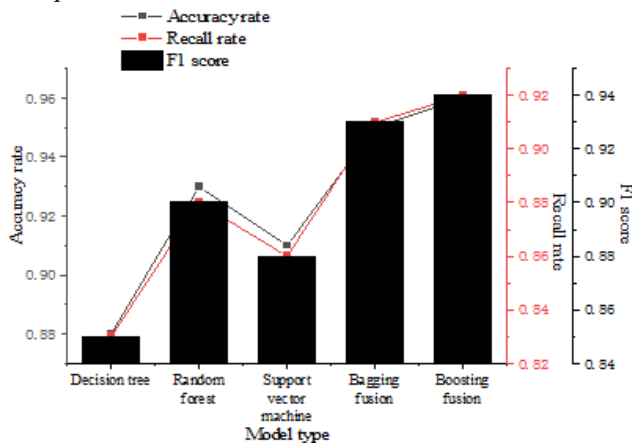


Fig. 1. Results of Model Fusion Experiment

D. Hyperparameter Optimization Experiment

1) *Experimental objective*: To optimize the hyperparameters of the model to improve performance.

2) *Experimental process*: Grid search and random search were used to search for the optimal combination of hyperparameters, and the effects before and after optimization were compared.

Experimental steps:

a) *Step 1. Data processing*: Collect a dataset for network intrusion detection, including relevant feature vectors and labels. The data preprocessing stage includes data purification, processing of missing values, handling of outliers, encoding and normalization of features.

b) *Step 2. Mode selection*: Select a machine learning algorithm, such as random forest, neural network, etc., as the basic model.

c) *Step 3. Definition of hyper parameter space*: Determine the hyper parameter and its possible size. Choose hyperparameters based on the selected model type.

d) *Step 4. Hyperparameter optimization method*:

i) *Grid search*: Traverse all possible hyperparameter combinations within a determined hyperparameter space, and then use mutual confirmation method to evaluate the effectiveness of each hyperparameter combination. On this basis, select the combination with the best performance as the final hyper parameter.

ii) *Random search*: Randomly select a set of hyper parameter combinations from the defined hyper parameter space, and select the optimal hyperparameter combination from them.

e) *Step 5. Model training and evaluation*: Train the model using the optimal hyperparameter combination and evaluate its performance.

f) *Step 6. Performance comparison* is achieved by comparing the performance without hyperparameter optimization.

The hyper parameter optimization method is unoptimized, with an accuracy of 0.89; the hyper parameter optimization method is grid search optimization; the hyper parameter optimization method is random search optimization, with an accuracy of 0.93. The results of the hyper parameter optimization experiment are shown in Figure 2.

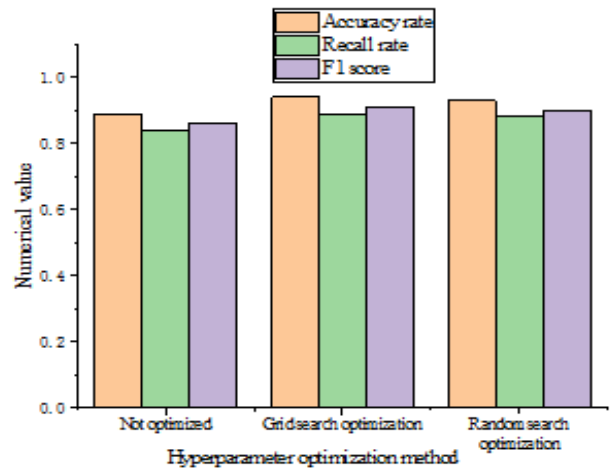


Fig. 2. Results of hyperparameter optimization experiment

E. Real-time Performance Testing

1) *Experimental objective*: To evaluate the performance of the system in a real-time environment.

2) *Experimental part*: Simulate the constructed model and verify its real-time performance.

Experimental steps:

a) *Step 1. Model preprocessing*: Utilize historical data from information security network intrusion detection systems and store it in a database.

b) *Step 2. Real-time data collection*: Build a data collection model to collect network information in real-time, such as network packets, logs, etc.

c) *Step 3. Data preprocessing*: First, perform feature extraction, encoding transformation, and normalization on the data.

d) *Step 4. Real-time detection*: Based on these detection results, the established model is preprocessed to effectively detect information security network intrusion situations.

e) *Step 5. Performance monitoring*: record time period, detection accuracy, false alarm rate, etc.

The real-time performance test results are shown in Figure 3. The accuracy rate of 08:00-09:00 is 0.92, the false positives rate is 0.05, and the average detection time is 50 milliseconds; the accuracy rate of 09:00-10:00 is 0.90, the false positives rate is 0.06, and the average detection time is 55 milliseconds; the accuracy rate of 10:00-11:00 is 0.91, the false positives rate is 0.05, and the average detection time is 48 milliseconds.

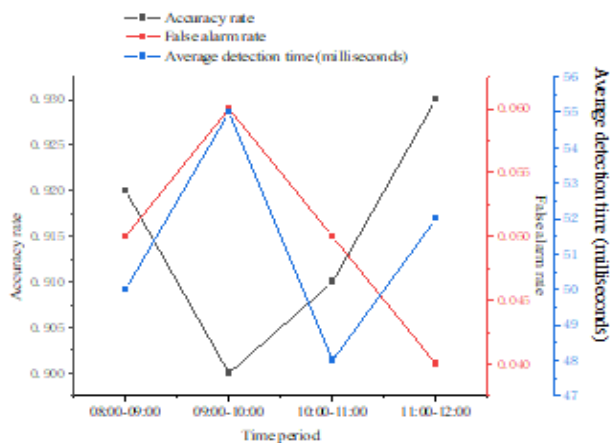


Fig. 3. Real-time performance test results

V. CONCLUSIONS

Traditional intrusion detection techniques are based on rule detection and feature engineering, relying on manually defining rules and selecting features, which cannot effectively respond to the evolution of intrusion behavior. Nowadays, with the development of machine learning, it has been able to automatically mine the rules and patterns contained in data, achieve automatic recognition and response to network attacks, and thus improve the accuracy and efficiency of IDS. This article mainly studies a machine learning based network intrusion detection method. This article intends to conduct experimental research on the above research results to verify their correctness and advantages, and provide technical support for improving network security defense capabilities. In the future, intelligent algorithms can be further optimized for real-time detection of network intrusions.

AUTHOR CONTRIBUTIONS

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